# Biswajit Pain

Systems Development Engineer / Platform Engineer

Berlin, Germany • +49 151 24865920 • biswajit.pain@outlook.com

## PROFILE

Systems Development Engineer with 12+ years building and operating production cloud and distributed-storage infrastructure across AWS, Delivery Hero, Sixt, and VMware. Hands-on experience with Ceph (RBD, RGW, CephFS), Rook on Kubernetes, OpenStack (Cinder, Swift, Nova), and VMware vSAN / vSphere storage, delivered in both production and internal-platform environments. Day-to-day engineering in Go, Python, and Bash across AWS, GCP, and private datacenters; deep troubleshooting through Linux, networking, and the storage layer — from block device and filesystem behavior up to S3 / object gateway semantics.

## TECHNICAL SKILLS

**Languages:** Go, Python, Bash, C, Java

**Kubernetes & Containers:** Kubernetes, Helm, container runtimes, CRDs / operators, microservices

**Cloud & IaaS:** AWS (EC2, S3, Route53, RDS, CloudFront, CloudWatch, SES, SNS, IAM), GCP, VMware IaaS

**Ceph & Distributed Storage:** Ceph RBD, RGW, CephFS; Rook operator on Kubernetes; CRUSH maps, placement groups, BlueStore OSDs, erasure coding, replication factors, pool and bucket design; cluster recovery, scrub / backfill tuning, RGW multi-site

**OpenStack:** Cinder (block), Swift (object), Nova (compute), Keystone, Neutron; Ceph-backed Cinder and Glance, OpenStack + Ceph integration patterns

**Storage:** VMware vSAN, vSphere storage, enterprise storage (IBM, Hitachi), NAS, S3 / object storage, block volumes, replication, failover

**Data:** MySQL, PostgreSQL, Kafka, Elasticsearch, data migration

**Infra as Code / CI/CD:** Terraform, Chef, Puppet, Jenkins, Git-based pipelines

**Observability:** OpenTSDB, Graylog, Fluentd, Logstash, rsyslog, cAdvisor, Nagios

**OS / Networking:** Linux (RHEL, Ubuntu), performance tuning, TCP/IP, VPN (hub-and-spoke), HAProxy, load balancing

**Web Frameworks:** Django, Flask

## EXPERIENCE

**Lead Systems Development Engineer***Oct 2022 – Present*

**Amazon Web Services (AWS)***Berlin, Germany*

***Database service — region build and storage layout***

- Owned the redesign of the region build process for a managed database service with a large distributed storage footprint, deployed across AWS regions.
- Rebuilt the provisioning pipeline so new regions come up with correct control-plane topology, storage layout, replication configuration, and networking without manual intervention.
- Shortened time-to-first-customer in a new region and removed a class of bootstrap failures caused by race conditions in the storage and control-plane initialization flow.

***Sovereign Cloud — Berlin engineering org***

- Part of the founding engineering group setting up AWS European Sovereign Cloud infrastructure in Berlin.
- Ran hundreds of technical interviews (systems design, coding, service deep-dive) to build the local engineering bench for a regulated, isolated cloud region.
- Contributed to early technical decisions on service isolation, operator access, and data-residency boundaries.

***Game orchestration platform***

- Engineer on the platform that orchestrates game server fleets for one of the largest game-hosting services on AWS.

- Worked on scheduling, capacity, and low-latency session placement across regions.

*Team & operations*

- Technical lead for 10 engineers split across Berlin and a second continent; own architecture reviews, design docs, and on-call quality for the team's services.

**Staff Systems Engineer***Apr 2020 – Sep 2022*

**Delivery Hero SE***Berlin, Germany*

*Zero-downtime MySQL migration on GCP*

- Led migration of a large production MySQL estate to Google Cloud SQL for a globally used ordering platform.
- Designed the cutover using replication, read-only windows, and application-level dual-write verification so there was no visible downtime for restaurants or customers.
- Built pre- and post-migration consistency checks (row counts, checksum-based diff on critical tables) to prove correctness before committing to the new primary.

*Fintech payment reliability*

- Debugged and fixed recurring failures in the payment processing path: idempotency gaps, duplicate-charge edge cases, and timeouts against external PSPs.
- Added retry/backoff, circuit breakers, and clearer reconciliation jobs; cut payment-related incident volume and improved settlement accuracy.

*Internal tooling*

- Wrote internal CLIs and dashboards (Python + Grafana) to speed up incident triage across the distributed payment services.

**Manager III, Site Reliability Engineering***Apr 2018 – Mar 2020*

**Sixt Research & Development***Berlin / Munich, Germany*

*Hydrant — infrastructure automation platform*

- Led Hydrant: an internal platform that automated provisioning of AWS accounts, VPCs, IAM baselines, and standard service stacks for product teams.
- Replaced hand-rolled tickets and ad-hoc Terraform usage with a self-service flow; cut new-environment setup from days to under an hour.
- Stack: Terraform, Python, Jenkins, AWS Organizations.

*PCI DSS–compliant payment platform on AWS*

- Designed and deployed the payment platform against PCI DSS requirements: network segmentation, restricted IAM, centralized logging, encryption at rest and in transit, audit trails.
- Drove the audit with external assessors and passed compliance on first attempt.

*Team leadership*

- Managed 7 SREs across Munich and Bangalore; ran hiring, on-call rotations, and roadmap planning across the two sites.

**Member of Technical Staff II***Nov 2016 – Mar 2018*

**VMware Software India Pvt. Ltd.***Bangalore, India*

*Ceph storage for multi-cloud IaaS — production and internal*

- Designed and operated Ceph clusters providing block (RBD), object (RGW / S3-compatible), and file (CephFS) services backing the multi-cloud IaaS platform used by internal product teams and customer-facing workloads.
- Sized and built the CRUSH map and pool layout (replicated and erasure-coded pools, separate device classes for SSD and HDD tiers) to match durability and latency targets for different workload classes.
- Tuned BlueStore OSDs, placement group counts, scrub schedules, and recovery/backfill throttles to keep client IO predictable during node loss and rebalance events.
- Operated RGW for S3-compatible object storage: bucket lifecycle policies, user and quota management, and multi-site replication between sites.

*Rook operator on Kubernetes*

- Deployed and operated Rook to run Ceph on Kubernetes for internal platforms, automating cluster bring-up, OSD lifecycle on node churn, and version upgrades through the operator.
- Built StorageClasses and CSI bindings so product teams could consume RBD volumes and CephFS shares as first-class Kubernetes PersistentVolumes.
- Wrote runbooks and automation around the Rook CRDs (CephCluster, CephBlockPool, CephObjectStore) for day-2 operations: capacity expansion, pool changes, and failure recovery.

*OpenStack + Ceph integration*

- Integrated Ceph with OpenStack services — RBD as the backend for Cinder (block) and Glance (image), Swift-compatible object storage via RGW — so tenants got unified storage across compute, image, and object APIs.
- Worked across Nova, Cinder, Neutron, and Keystone to debug cross-component issues (volume attach races, auth token failures, network policy and QoS at the storage network layer).
- Reviewed tenant isolation, rate-limiting, and quota enforcement for multi-tenant shared storage.

*vSAN and vSphere storage*

- Ran vSAN clusters and vSphere storage policies in parallel with the Ceph environments; compared vSAN-based and Ceph-based backends on performance, operational cost, and failure modes to guide platform direction.

*Platform engineering — microservices and CI*

- Broke out services from a large monolithic test/CI system into independently deployable microservices with clear API boundaries; introduced service-level health checks and per-service deployment pipelines.

**Production Engineer***Jan 2016 – Oct 2016*
**ANI Technologies (Ola)***Bangalore, India*

*HAProxy log analytics pipeline*

- Built the HAProxy log pipeline using rsyslog, Kafka, Logstash, and Elasticsearch for real-time visibility into traffic patterns and error rates across the fleet.
- Used the pipeline to identify slow upstreams and misbehaving clients that had previously been invisible.

*Microservice infrastructure*

- Contributed to the platform supporting the company's rapid user growth: service discovery, deploy tooling, and base images.

**DevOps Engineer***Apr 2014 – Dec 2015*
**Knowlarity***Bangalore, India*

*Multi-datacenter VoIP telephony infrastructure*

- Designed and built the multi-datacenter VoIP telephony infrastructure powering the company's cloud telephony product across India and SEA.
- Built the active-active datacenter layout with SIP trunk termination across sites, media server pools, and carrier-level redundancy so call traffic survived single-DC loss.
- Owned the failover platform end-to-end: DNS, load balancer, Asterisk/FreeSWITCH media nodes, and DB replica cutover runbooks.

*Complex hub-and-spoke network*

- Designed and rolled out a hub-and-spoke network connecting multiple datacenters, office sites, and carrier interconnects with redundant IPsec tunnels and dynamic routing.
- Replaced an unstable full-mesh VPN setup; resolved persistent inter-site routing loops, asymmetric paths, and fragmentation issues that had been causing dropped calls and signaling failures.
- Tuned OSPF/BGP route preferences and failover timers so spoke sites re-converged on the surviving hub within seconds of a link loss.

**Assistant Systems Engineer / Trainee***Jul 2013 – Mar 2014*
**Tata Consultancy Services (TCS)***Kolkata, India*

- Delivered infrastructure automation work including storage virtualization tasks for a client engagement.
- Led a team of five during the Initial Learning Program.

## EDUCATION

**B.Tech, Computer Science and Engineering** — Kalyani Government Engineering College, India (2013)